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SECTION 02
GROUP BIII

PROJECT:
IMPROVING U.S. AIRLINE OPERATIONS THROUGH FLIGHT DELAY ANALYSIS
PHASE 2

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1.0 Introduction

The Phase 2 workflow documentation for the Business Intelligence (BI) project on U.S. airline operations is presented in this paper. This phase’s goal is to use Alteryx Designer to design and construct a comprehensive data preparation and integration pipeline that unifies three raw datasets into an output that is suitable for analysis. The final datasets will be used in Phase 3 for dashboard construction and display.

2.0 BI Architecture Overview

Data input, data preparation and data output are the three layers of the workflow’s architecture. Before being combined via join and union procedures, each of the three source datasets is loaded separately and goes through type conversion, cleansing, filtering and deduplication. Three output files with various analytical uses are produced by the end pipeline.

2.1 Input Datasets

Table 1: Input Datasets Description

Dataset	File Name	Purpose
U.S. Flights 2023	US_flights_2023.csv	Main flight records: delay times, departure/arrival info, delay causes, aircraft details
Cancelled & Diverted 2023	Cancelled_Diverted_2023.csv	Cancellation and diversion flags per flight with Cancelled/Diverted columns
Aircraft Registry	maj us flight – January 2024.csv	Aircraft reference data: manufacturer, model and aircraft age

3.0 Alteryx Workflow Description

There are 46 tools arranged in three simultaneous processing streams that come together at a single integration and output stage to form the workflow. The arrangement of the canvas is horizontal.

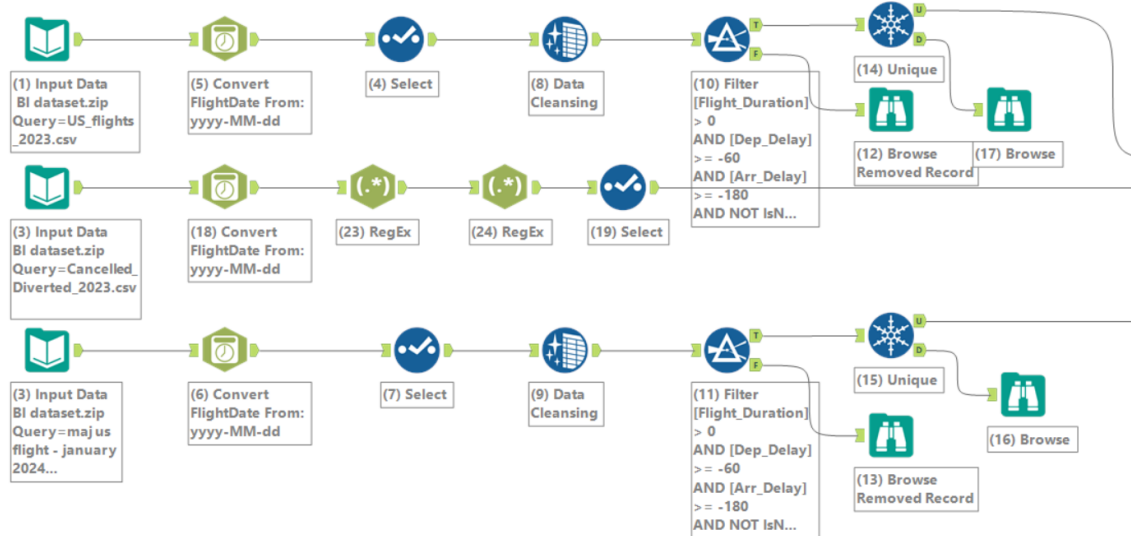


Figure 1: Data Cleaning, and Deduplication Process (Tools 1–17)

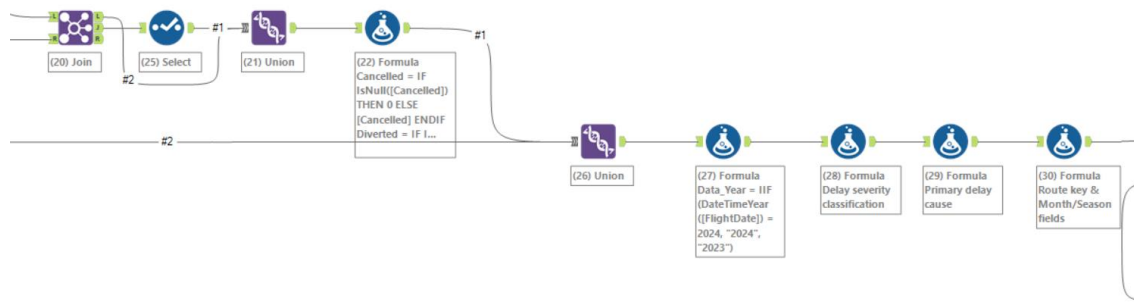


Figure 2: Alteryx Workflow – Data Integration and Feature Engineering: Part 1 (Tools 20–30)

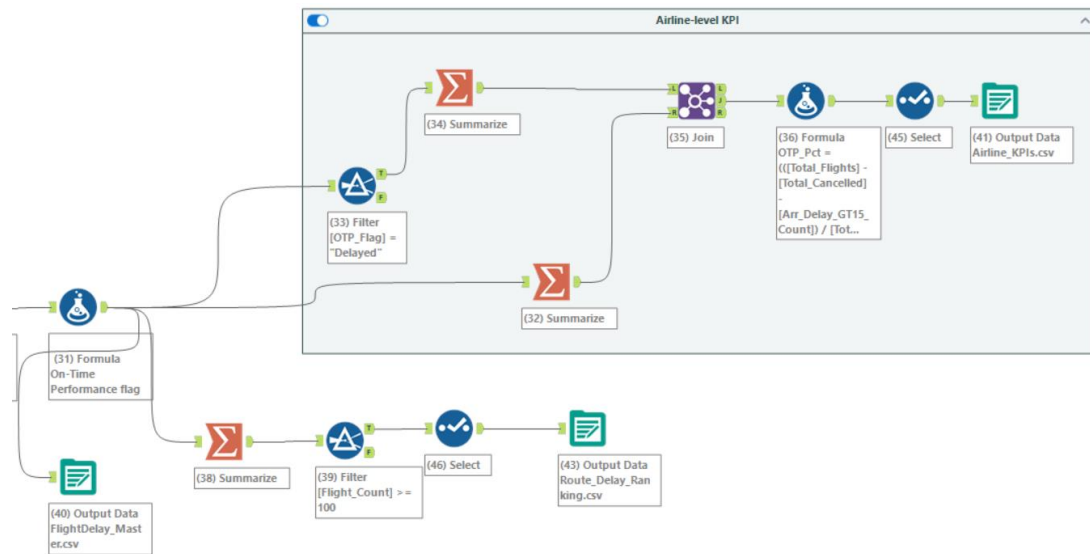


Figure 3: Alteryx Workflow – Feature Engineering: Part 2, Aggregation, and Output Generation (Tools 31–46)

The process of the workflow starts with loading all three dataset sources independently with the Input Data tool. The files are read using Latin-1 encoding and comma as the delimiter:

- US_flights_2023.csv – the main flight records dataset
- Cancelled_Diverted_2023.csv – cancellation and diversion flags
- maj us flight – January 2024.csv – aircraft registry reference date

3.1 Data Cleaning

Each of the 3 streams is treated separately for data cleaning prior to integration, which results in consistency and validity of all datasets as illustrated in Figure 1. The following steps shows what has been done in this workflow:

1. Data Type Conversion

Date Time tool used to convert FlightDate string field into a proper DateTime format. This is required because data functions require a DateTime type and not a string type at the downstream level of the tools. The converted field will be temporarily named FlightDateTime_Out in the next Select tool, FlightDate.

2. Data Cleansing

Data cleansing process is applied using the **Data Cleansing Tool**. This includes:

- Removing leading and trailing spaces
- Standardizing text values to uppercase
- Replacing empty strings with null values

Trimming whitespace from the data is particularly important for Tail_Number, as any trailing space would result in the join failing quietly, discarding good data from the merged data set.

3. Correct format inconsistencies in cancelled and diverted data

The 'Cancelled' and 'Diverted' columns in Cancelled_Diverted_2023.csv were stored as floating-point values (eg. 1.0, 0.0). Direct integer casting would fail because of the decimal suffix. Two **RegEx tools** using the replacement pattern '\.0\$' strip the trailing '.0' from both fields before the Select tool casts them to Int16.

4. Record Filtering

Filter Tools used to remove invalid or corrupted data. Flights with:

- Negative or zero flight duration
- Missing airport or airline information
- Extreme delay values

are filtered out to ensure only valid flight records are retained for analysis. Records which pass any of the conditions are forwarded to the Browse tool 12, 13 with a "Removed Record" annotation to be reviewed.

5. Deduplication

Duplicate records are eliminated by a composite key of five fields using **Unique tools**: 'FlightDate', 'Tail_Number', 'Dep_Airport', 'DepTime_label', and 'Arr_Airport'. This combination is unique to a single flight leg. Records that are duplicated are sent to Browse tool 16, 17 for review. The output of the Unique tool is directly used in the integration stage later.

6. Null Filling Post-Integration

This step fills in the blanks after the Union tool as shown in Figure 2, in the 'Cancelled' and 'Diverted' columns in the US Flights stream with the matching cancelled and diverted statuses for flight that had no record in the Cancelled/Diverted stream. These nulls will be replaced with 0.

3.2 Data Integration

After cleaning, the cleaning results are integrated into two stages in this way as shown in Figure 2.

Stage 1 – Join Operation

The "US Flight" dataset is first joined with the "Cancelled_Diverted" dataset using a **Join tool** with multi-field key consisting of:

- Flight Date
- Tail Number
- Origin Airport
- Destination Airport
- Departure Time Label

A composite key is used instead of only Tail Number because a single aircraft can operate multiple flights in a day. This ensures accurate matching at the flight-level granularity. The next tool in the series, **Select tool 25**, is used to correct the duplicate column prefixes that were generated by the join by stripping them off again to restore their original names.

Stage 2 – Union with Aircraft Dataset

The integration is finished with two **Union tools** in ByName mode. The records from the US Flights dataset (left output from tool 20) are combined with the cleaned join output from tool 25 ensuring flights without a cancellation record are still retained in the final dataset. Then, tool 26 merges the output from tool 21 with the cleaned Aircraft Registry stream (tool 15), producing a single unified dataset containing flight performance, cancellation status and aircraft characteristics for every record.

3.3 Data Transformation

The integrated and cleaned data exits the Union tool followed by five **Formula tools** that add derived analytical fields and then by aggregation steps that result in the output files as shown in Figure 3. Steps below showing the use of Formula Tools:

1. Year Labeling (Formula Tool 1)

Each record is labeled with which year it is, using a 'IFF(DateTimeYear([FlightDate]) = 2024, "2024", "2023")'. This enables year-based filtering and comparison in the dashboard as the aircraft registry file contains data for January 2024 flights and the main dataset contains data for 2023.

2. Delay Severity Classification (Formula Tool 2)

Flights are categorized into:

- On Time (≤ 0 min)
- Minor Delay (1–15 min)
- Moderate Delay (16–45 min)
- Severe Delay (46–120 min)
- Critical Delay (>120 min)

This categorical variable allows the use of distribution and stacked bar charts to look at the distribution of delay severity by airline, route and season.

3. Primary Delay Cause (Formula Tool 3)

This field is created by comparing the five delay component columns (Carrier, Weather, NAS, Security, Late Aircraft) using a series of nested ELSEIF statements. Flight with a 'Dep_Delay' of ≤ 0 will be classify as "No Delay". This field allows cause-breakdown analysis without having to compare five numeric columns on the dashboard by hand.

4. Route, Seasonal and Aircraft Features (Formula Tool 4)

A route field is generated by combining origin and destination airports (e.g., ATL → LAX), enabling route-level analysis. 'Month_Num' is a variant of 'DateTimeMonth()' that returns the numeric value of the month. Season defines each month as either a Winter (Dec-Feb), a Spring (Mar-May), a Summer (Jun-Aug), or a Fall (Sep-Nov). 'Is_Weekend' is a Boolean function that is true if 'Day_Of_Week' is in (6, 7) or is Sunday or Saturday.

5. Aircraft Age Grouping (Formula Tool 4)

Aircraft are grouped into:

- New (0–5 years)
- Mid-life (6–15 years)
- Aging (16+ years)

This helps analyse whether older aircraft contribute more to delays.

6. OTP Flag and Total Delay Minutes (Formula Tool 5)

‘OTP_Flag’ indicates if the flight was "On Time" or "Delayed" according to ‘Arr_Delay’ ≤ 15 , which is the standard on-time definition provided by the FAA. ‘Total_Delay_Minutes’ is adding all positive delay components, adding value to the end, but wrapping it with $\text{MIN}(\text{MAX}(0, \text{value}), \text{MAX}(0, \text{value}))$ to exclude negative values which are called early-segment absorption.

7. Route-Level Aggregation

This step performs the average delay by minutes, the number of flights and average weather and carrier delays by route using the **Summarize tool** (Tool 38). Next a **Filter tool** (Tool 39) filters out routes with fewer than 100 flights to make sure there is a statistical significance for the analysis. The delay averages are rounded to two decimal places before they are output by a **Select tool** (Tool 46).

8. Airline-Level KPI Aggregation

Two Summarize tools are used here to provide the computation of KPIs grouped by ‘Airline’, ‘Data_Year’ and ‘Season’. The overall metrics that Tool 32 provides are average delays, total flights, total cancellations, diversions, and total delay minutes. Tool 33 first filters for delayed records only (OTP_Flag = "Delayed"), and Tool 34 counts them to produce ‘Arr_Delay_GT15_Count’. Tool 35 combines these two summaries into one place, thus making all KPI columns available. Tool 36 then calculates the final ‘OTP_Pct’ as follows: $(\text{Total_Flights} - \text{Total_Cancelled} - \text{Arr_Delay_GT15_Count}) / \text{Total_Flights} \times 100$. The decimal fields are formatted in Tool 45 before the output.

4.0 Challenges Faced

During the design of Alteryx workflow, there are several challenges encountered across the data integration and output stages. It needs careful diagnosis and targeted corrective actions to ensure the accuracy and integrity of final output datasets.

4.1 Challenge 1: Duplicate Column Naming After Join

During the integration of US Flights and Cancelled & Diverted dataset via Join tool (Tool 20), Alteryx automatically prefixed all matching column names from the Right input with “Right_”. Thus, producing fields such as Right_FlightDate and Right_Tail_Number. This caused schema conflicts in downstream tools and triggered field-not-found warnings in the next, Select tool (Tool 25). Therefore, to resolve the problem, Select tool (Tool 19) was configured on the DS2 branch prior to the Join to retain only the five join key fields and the two payload columns, Cancelled and Diverted. Then, Select tool (Tool 25) was used post-Join to rename the remaining prefixed columns back to their original names, restoring a clean 26-column schema for downstream processing.

4.2 Challenge 2: Loss of Decimal Precision in Aggregated Delay Fields

Aggregated delay averages in Airline_KPIs.csv and Route_Delay_Ranking.csv initially produced while numbers rather than decimal values. This occurred because numeric fields such as Dep_Delay and Arr_Delay were not explicitly initialized during ingestion, causing Alteryx to default them to integer types in Select tools (Tools 4 and 7). Since precision was lost, downstream ROUND expressions were unable to recover fractional values. The issue was solved by explicitly casting all eight numeric delay and duration fields to Double within the Select tools at the point of ingestion to ensure full decimal precision was preserved throughout the whole

pipeline. Lastly, the output fields in Select tools (Tools 45 and 46) were additionally set to FixedDecimal to enforce consistent two-decimal-place formatting across all output files.

4.3 Challenge 3: Unicode Encoding Error in Route Field Output

The Route field was constructed in Formula tool (Tool 30) using the Unicode arrow character → (U+2192) and caused the Output Data tool (Tool 40) to generate a character conversion error on every record. This was caused by the output was configured with Latin-1 encoding (code page 28591), which does not support Unicode characters beyond 256 code points. The error has entirely blocked the generation of FlightDelay_Master.csv. Besides, the Write BOM option was enabled, which adds a hidden character sequence at the start of the file that fails the recognition of first column name when the output CSV is loaded into visualization tools in Phase 3. Both issues were tackled by changing the code page to UTF-8 (65001) and unchecking Write BOM across all three Output Data tools (Tools 40, 41 and 43) to ensure a consistent and clean encoding for all output files.

5.0 Results

Three output files were created once the Alteryx workflow was successfully completed as shown in Table 1.

Table 2: Output file description

Output File	Key Summary
FlightDelay_Master.csv	7,244,503 records (15 airlines, 4 seasons). The dashboard's primary dataset has derived fields including Delay_Severity, OTP_Flag, Route, and Aircraft_Age_Group. About 1.71 GB in size.
Airline_KPIs.csv	60 records for airline seasons. Republic Airways performed the best (3.29 minute delay, 87.13% OTP). The worst are Frontier (68.70% OTP) and JetBlue (24.42 min delay). Fall has the best delay (8.15 minutes), while summer has the worst (17.70 minutes).
Route_Delay_Ranking.csv	9,250 routes (more than 100 flights). The best route is OTZ →ANC (-8.38 min, Alaska Airlines). RNO→JFK (91.11 min, JetBlue) is the worst. JetBlue has the longest average delay (25.15 minutes), followed by Frontier (22.14 minutes).